

CURRICULUM BOOK
ACADEMIC SESSION 2024-25
CLASS-VII
SUBJECT- SCIENCE
CHAPTER: 3: HEAT

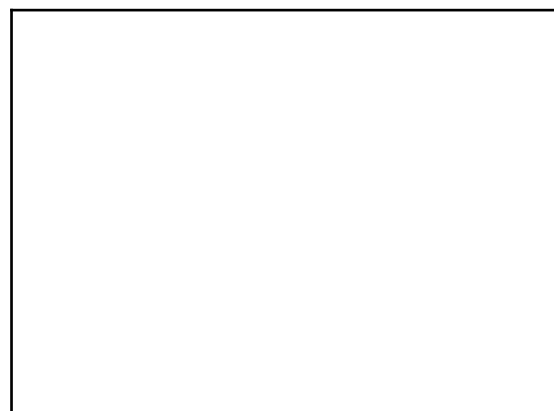
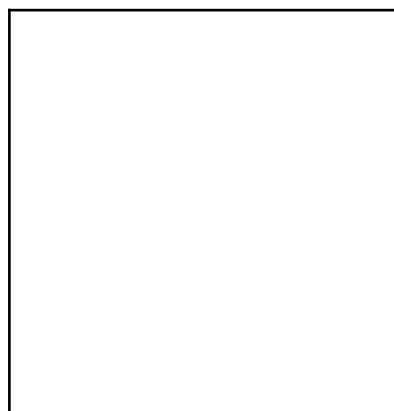
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OVERVIEW (MSP)

ACADEMIC SESSION 2024-25

Facilitator's name	Grade	Subject	Chapter name	Date
	VII	Science	Heat	

Content	Concept	Knowledge
- Hot and cold - Measuring temperature - Laboratory thermometer - Transfer of heat - Kinds of clothes we wear in summer and winter	- Concept of heat and cold - Temperature definition - Measurement of temperature - Measuring temperature of our body with a clinical thermometer - Measuring temperature of water with a laboratory thermometer - Conductors and Insulators - Conduction - Convection - Radiation - Sea breeze and Land breeze - Choice of clothes in various seasons	- Our sense of touch is not always a reliable guide to the degree of hotness of an object. - Temperature is a measure of the degree of hotness of an object. - Thermometer is a device used for measuring temperatures. - Clinical thermometer is used to measure our body temperature. The range of this thermometer is from 35°C to 42°C. For other purposes, we use laboratory thermometers. The range of these thermometers is usually from -10°C to 110°C. - The normal temperature of the human body is 37°C. - The heat flows from a body at a higher temperature to a body at a lower temperature. There are three ways in which heat can flow from one object to another. These are conduction, convection and radiation. - In solids, generally, the heat is transferred by conduction. In liquids and gases the heat is transferred by convection. No medium is required for transfer of heat by radiation.



- The materials which allow heat to pass through them easily are conductors of heat.
- The materials which do not allow heat to pass through them easily are called insulators.
- Dark-coloured objects absorb radiation better than the light-coloured objects. That is the reason we feel more comfortable in light-coloured clothes in the summer.
- Woolen clothes keep us warm during winter. It is so because wool is a poor conductor of heat and it has air trapped in between the fibres.

Competences
3C Competence: Communicator
Milestone: Communication Skill

Character
3C Character: Participative
Milestone: Thinking Skill
Sustainable Development Goals (SDG): Sustainable Development Goal (SDG) 7: Affordable and Clean Energy.

Action
Identify the displays of temperature and weather forecast set up in your locality.

Sports Education Program	Many injuries occurred while playing can be treated by using heat bag fermentation.
Visual Arts & Performing Arts	Many arts and crafts can be manufactured by using heating techniques.

Integration

Mathematics: There are two mathematical rules governing the relationship between heat flow and temperature change: ‘Heat flow is directly proportional to spatial rates of change in temperature distribution’. ‘The rate of temperature change at any point is proportional to the rate of heat flow into that point.’

Social Science: Integrating to Social where we come across renewable energy.’

SESSION TRACKER
ACADEMIC SESSION 2024-25

Elements	Subject- Science
Chapter Name	Heat
No. of periods allotted	10
Sub-topics to be taught	1. Hot and cold 2. Measuring temperature 3. Laboratory thermometer 4. Transfer of heat 5. Kinds of clothes we wear in summer and winter
Teaching Strategies	PPT, Video, Task Sheets, Worksheets, Chapter tour, Heads together, Video & questioning, Investigation, Think- group-write-share, Brainstorming, Jigsaw and read aloud strategies.
Field trips <i>(If any) (Yes/ No)</i>	NA
Lab activities <i>(If any) (If yes, please name)</i>	1. To demonstrate that some materials are good conductors of heat while some are insulators. 2. To demonstrate the transfer of heat by conduction. 3. To study the process of transfer of heat in liquids by convection. 4. To measure the temperature of hot water by using a laboratory thermometer.

	5. To study the thermal expansion in solids.
Integration of technology by the facilitator/ students	YES
Domain/Levels of Bloom's Taxonomy targeted	Remembering, Understanding, Applying, Analysing, Evaluating and Creating.
Differentiated Learning Multiple Intelligences planned to be incorporated	<u>(To be filled in by the school subject teacher after diagnosis assessment)</u>

- ☐ Co-scholastic integration consists of performing arts, visual arts and physical education.
- ☐ Use of technology by the facilitator and the students is mandatory.
- ☐ The report must be submitted 2 weeks before the beginning of the chapter.
- ☐ Carefully study the format before filling and submitting.
- ☐ Columns may be added by the facilitator if required. However no column can be deleted.

Facilitator signature	Coordinator signature	Principal signature
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RESOURCE LIST
ACADEMIC SESSION 2024-25
Chapter-03 Heat

Sl. No.	Description/ Details of Resource	File Name	Annexure No.
1	Worksheet 1	Grade 7_Science_Ch 3_Heat_WS1_2024-25	Annex I.1
2	Worksheet 2	Grade 7_Science_Ch 3_Heat_WS1_2024-25	Annex I.2
3	Transfer of heat	Grade 7_Science_Ch 3_Heat_PPT1_2024-25	Annex II.1
4	Introduction	https://www.youtube.com/watch?v=xGKg3TSO4v8	Annex III.1
5	Measuring temperature	https://www.youtube.com/watch?v=J157oziu3zQ	Annex III.2
6	Why is a laboratory thermometer not used to check body temperature?	https://www.youtube.com/watch?v=EQvATCqVB9o	Annex III.3
7	Why do woollen clothes keep us warm?	https://www.youtube.com/watch?v=DWaWocvbpQI	Annex III.4
8	Task sheet-1	Grade 7_Science_Ch 3_Heat_TS1_2024-25	Annex IV.1
9	Task sheet-2	Grade 7_Science_Ch 3_Heat_TS2_2024-25	Annex IV.2
10	Task sheet-3	Grade 7_Science_Ch 3_Heat_TS3_2024-25	Annex IV.3

THE ESSENTIAL ELEMENTS OF SESSION PLAN

ANTICIPATORY SET	INSTRUCTIONAL OBJECTIVE PURPOSE	❖ Get learner focused ❖ To be used at - Start of lesson ▪ Interventions ▪ At a transition
CORE INSTRUMENTAL ELEMENT	PROVIDE INFORMATION MODELLING THE INFORMATION OR PROCESS MONITOR AND ADJUST (check for understanding)	❖ Organise basic information ❖ Simplest and clearest ❖ Respond to queries ❖ Highlight critical attributes ❖ Avoid controversies in first model ❖ Accurate and unambiguous ❖ Finally introduce non-exemplars ❖ Signalled answers ❖ Choral responses ❖ Sample individual responses

	GUIDED PRACTICE (during lesson)	❖ Massed ❖ Distributed - How much/ how long/ how often
CLOSURE	INTERNALISE LEARNING	❖ Learners must summarise learning ❖ Bridge to future learning ❖ Exercise opportunity to monitor
INDEPENDENT PRACTICE	SELF STUDY AND HOME ASSIGNMENTS	❖ Correction ❖ Feedback

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	1	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Introduction
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to identify the importance and usage of heat in daily life.
Instructional Resource(s)	NCERT text book, video-1 (Refer Annex III.1)
Facilitation Procedure (How / Strategies/Activities/ Tools)	CHAPTER TOUR STRATEGY is used to facilitate this session. MOTIVATIONAL and CLASS ACTIVITY: (25 min) 1. Facilitator initiates the session by posing questions to students to recall their prior knowledge on heat concepts. (RECALL) a. If you leave a hot coffee on the table for a while, does it heat up or cool off? b. If you leave a cold drink on the table for a while, does it heat up or cool off? c. How to measure the hotness or coldness of a body? d. Can you lift a hot vessel without being hurted? e. When you are suffering from fever, how are you going to measure your body temperature? f. Notice the change when you mix some cold water in a small tank of hot water.

- g. *People, who stay near coastal areas, feel more hotness during sunny days and more coldness during nights when compared to others. Do you know why?*
- h. *What type of clothes do you prefer to wear in the winter season and why?*
- i. *What type of clothes do you prefer to wear in the summer season and why?*
- j. *In which of the following cases, you feel hotness when you touch it. a) Iron handled pan, b) Wooden handled pan?*



2. Facilitator divides students into 3 groups and instructs them to write down the above questions in their notebook. (Time allocated: 5-6 min.)
3. Students in groups apply their previous knowledge to answer the given questions by discussing with group members. (Time allocated: 10-12 min.) (APPLY)
4. After allotted time, the facilitator calls one member from each group to read out their answers. Some of the guesses may be correct.
5. Now, the facilitator plays a video-1 (Refer Annex III.1) in which heat and its usage/importance is explained. (Time allocated: 6-7 min.)

REINFORCEMENT: (10 min)

6. Further, the facilitator introduces the respective phenomena/concept by using the above given questions so that the chapter tour is done and he/she continues saying that the topics will be explained in detail in coming sessions. (Time allocated: 4-5 min.)
7. Facilitator summarises the session by inviting one of the students to give correct answers for posed questions and all the concept names that they are going to learn in this chapter. (Time allocated: 4-5 min.) (TELL)

	(Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Applying)	
Sustainable Development Goal Activity		
	SDG No.	SDG Name
	<i>SDG 6 Clean Water and Sanitation Refer to Grade7-Geography-Ch1-Environment_Curriculum Book_2024-25</i>	<i>SDG 6 Clean Water and Sanitation Refer to Grade7-Geography-Ch1-Environment_Curriculum Book_2024-25</i>
Independent Practice (Homework)	<u>Assertion and Reason Question:</u> Question 1: Assertion (A): Metal spoons get hotter faster than wooden spoons when placed in a hot liquid. Reason (R): Metals are good conductors of heat, whereas wood is a poor conductor of heat. Options: a) Both A and R are true, and R is the correct explanation of A. b) Both A and R are true, but R is not the correct explanation of A. c) A is true, but R is false. d) A is false, but R is true. Question 2: Assertion (A): A black surface absorbs more heat than a white surface when exposed to sunlight. Reason (R): Darker colours have higher emissivity, meaning they emit and absorb more thermal radiation compared to lighter colours. Options: a) Both A and R are true, and R is the correct explanation of A. b) Both A and R are true, but R is not the correct explanation of A.	

	c) A is true, but R is false. d) A is false, but R is true.
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	2	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Hot and cold
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to define temperature.
Instructional Resource(s)	NCERT text book, required components, Task sheet-1 (Refer Annex IV.1)

Facilitation Procedure (How / Strategies/Activities/ Tools)	HEADS TOGETHER STRATEGY is used to facilitate this session. MOTIVATION: (10 min) 1. Facilitator divides students into pairs and makes them sit as they are heading together. (Time allocated: 1-2 min.) 2. Facilitator initiates the session by instructing students to go through table 3.1 in the textbook and write observations in the textbook by discussing with the partner. (Time allocated: 3-4 min.) (DISCUSS) 3. Now, the facilitator asks one of the pairs to come forward and share their answers with all. (Time allocated: 3-4 min.) EVALUATION: (20 min) 4. Facilitator continues the session instructing students in pairs to do activity 3.1 in the textbook and write their observations in a notebook. (Time allocated: 8-9 min.) 5. Further, the facilitator introduces the concept of 'hot and cold' and defines that the degree of hotness and coldness is called 'temperature' by collecting observations from students on activity 3.1 and the textbook (Time allocated: 6-7 min.) 6. Facilitator also adds that 'thermometer' is the instrument used to measure the temperature of a body and students are supposed to write key points. (Time allocated: 2-3 min.) REINFORCEMENT: (5 min) 7. Facilitator distributes a Task sheet-1 (Refer Annex IV.1) to each pair and gives some time to answer as a part of recalling. (Time allocated: 3-4 min.) (RECALL) 8. Facilitator picks students at random to read out their answers to summarise the session. (Time allocated: 1-2 min.) (Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Creating)
Independent Practice (Homework)	Case Based Question:

	Case: Trupty is conducting an experiment in her science class to understand heat transfer. She has three identical metal rods: one painted black, one painted white, and one left as bare metal. She places one end of each rod into a pot of boiling water and measures the temperature at the other end after 5 minutes. She notices that the black-painted rod's end is the hottest, the bare metal rod's end is slightly cooler, and the white-painted rod's end is the coolest. Questions: 1. Why is the black-painted rod's end the hottest? 2. Explain why the white-painted rod's end is the coolest. 3. What principle of heat transfer is demonstrated in Trupty's experiment?
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	3	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Measuring temperature
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to identify how the temperature of a body can be measured using a thermometer.
Instructional Resource(s)	NCERT text book, video-2 (Refer Annex III.2)
Facilitation Procedure (How / Strategies/Activities/ Tools)	VIDEO AND QUESTIONING STRATEGY is used to facilitate this session. MOTIVATION & CLASS ACTIVITY: (25 min) 1. Facilitator recapitulates the previous session by using a few questions and video. a. Define temperature. b. Name the instrument used to measure the temperature of a device. (Time allocated: 1-2 min.) 2. Students come up with correct answers and the facilitator divides students into groups of 4 each. (Time allocated: 2-3 min.) 3. Facilitator now plays a video-2 (Refer Annex III.2) (up to 2 min. 48 sec) on ‘temperature’ concept to connect to the present session. (Time allocated: 2-3 min.)

	4. From the above video, students improve their knowledge on measuring temperature. 5. Now, the facilitator introduces the concept of ‘measuring temperature’ by showing and explaining the parts (scale, range of degrees, bulb, and mercury) of laboratory and clinical thermometers. (Time allocated: 8-9 min.) <i>Note: Facilitator instructs students to note down key points and parts of thermometer in their notebook.</i> 6. Further, the facilitator plays another video-2 (Refer Annex III.2) (from 2:49 to 9 min.) to improve students’ knowledge on ‘measuring temperature’. (Time allocated: 6-7 min.) (IMPROVE) REINFORCEMENT: (10 min) 7. After watching the video, students in groups are asked to answer the following questions in a notebook. - Name the instrument used to measure the body temperature of humans? - What is present inside the glass bulb of a clinical thermometer? - What are the two scales present on a clinical thermometer? - What is the range of the thermometer in $^{\circ}\text{C}$? - What is the range of the thermometer in Fahrenheit? - Mention the normal temperature of the human body in $^{\circ}\text{C}$. - Under what temperature the mercury level should be before using? (Time allocated: 4-5 min.) 8. Facilitator instructs groups to exchange their answers among themselves to evaluate and compare and summarises sessions by telling correct answers (EVALUATE) (Time allocated: 5-6 min.) (Bloom’s Taxonomy Levels Targeted: Understanding, Evaluating and Creating)
Independent Practice (Homework)	Students will be able to answer the Exercises Q. No. 1 in the textbook.

Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes
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SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	4	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Measuring temperature
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to experiment with activities to measure temperature of different bodies.
Instructional Resource(s)	NCERT text book, Task sheet-2 (Refer Annex IV.2)
Facilitation Procedure (How / Strategies/Activities/ Tools)	TEAM INVESTIGATION STRATEGY is used to facilitate this session. CLASS ACTIVITY: (30 min)

	- Facilitator divides students into two teams and names them as A and B. (Time allocated: 2 min.) - Initially the facilitator demonstrates activity 3.2 by calling one student from any team. (Time allocated: 4-5 min.) <i>Note: Facilitator informs all students need not to worry if their body temperature is slight below or above 37°C</i> - Both teams are assigned to go through activities 3.2 and 3.3 in the textbook and write their observations as investigation reports in an A4 sheet writing all of their names and their respective body temperatures. (Time allocated: 22-23 min.) <i>Note: Facilitator has to bring A4 sheets and has to make sure that each student is recording observation by himself/herself or not during activity 3.2.</i> - Further, the facilitator provides some time to students to write all observations in their investigation reports and collects. (Time allocated: 2 min.) REINFORCEMENT: (5 min) - Facilitator summarises the session saying that the body temperature range of a normal human being is 35°C to 42°C and recalls the point of why only clinical thermometers are used for measuring body temperature. (Time allocated: 2-3 min.) (Bloom's Taxonomy Levels Targeted: Understanding, Applying and Evaluating)
Independent Practice (Homework)	Students will be able to answer Task sheet-2 (Refer Annex IV.2).
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	2 X 35 min.
Session Number	5 & 6	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Laboratory thermometer
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to explain the usage and measuring of laboratory thermometers.
Instructional Resource(s)	NCERT text book, video-3 (Refer Annex III.3), worksheet-1 (Refer Annex I.1)
Facilitation Procedure (How / Strategies/Activities/ Tools)	<i>(Note: The facilitators can discrete the session and conduct the activities into two different sessions in the laboratory)</i> THINK-GROUP-WRITE-SHARE STRATEGY is used to facilitate this session. MOTIVATION: (10 min) 1. Facilitator initiates the session by posing a few questions to recapitulate their prior knowledge. a. Name the two thermometers you have seen. b. What is the range of a clinical thermometer? c. What is the normal temperature of the human body? d. Is the temperature of every person being 37°C?

	e. Which one is preferable to measure the temperature of the human body, lab or clinical thermometer? f. Can we make use of clinical thermometers to measure the temperature of other objects also? (Time allocated: 2-3 min.) 2. Facilitator divides students into groups of 4 each and asks them to think about the questions by discussing with group members. (Time allocated: 3-4 min.) 3. Students are asked to write answers in their notebook and the facilitator picks students randomly to share answers with all of the above questions. (Time allocated: 4-5 min.) CLASS ACTIVITY: (50 min) 4. Now, the facilitator explains that the laboratory thermometer is used to measure the temperature of other objects and shows it to all classes, explaining clearly the parts and its range. 5. Facilitator demonstrates activity 3.4 in the textbook and instructs students to listen to precautions carefully. (Time allocated: 7-8 min.) 6. Further, students are asked to do activity 3.2 in groups and write their observations in a notebook. (Time allocated: 10-12 min.) 7. Facilitator picks three or four students to read out their observations and the facilitator explains if there are any variations in values. (Time allocated: 3-4 min.) 8. Facilitator calls group wise to the experiment table to observe activity 3.3 while he/she is explaining and instructs remaining pairs to have a reading in the textbook. 9. The above step is repeated until all the pairs are done with the activity. (Time allocated: 12-13 min.) 10. Facilitator invites any one student to give a conclusion on activity 3.3 and corrects it if necessary. (CONCLUSION) (Time allocated: 2-3 min.) <i>Note: Expected conclusion is temperature must be read while the thermometer is in water.</i> REINFORCEMENT: (10 min) 11. Now, the facilitator plays a video-3 (Refer Annex III.3) and asks students to write down important points. (Time allocated: 2-3 min.) 12. Facilitator summarises the session by revising key points from the video and about laboratory thermometers and activities done in the same session. (Time allocated: 8-9 min.) Note: Students will be allowed to get individual experience during their lab period.
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	(Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Analysing)
Independent Practice (Homework)	Students will be asked to answer worksheet-1 (Refer Annex I.1)
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER - 3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	7	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Transfer of heat
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to explain how transfer of heat is occurring in different methods.
Instructional Resource(s)	NCERT text book, PPT1 (Refer Annex II.1)

Facilitation Procedure (How / Strategies/Activities/ Tools)	BRAINSTORMING STRATEGY is used to facilitate this session. MOTIVATION: (7 min) - Facilitator divides students into groups of 3 and initiates sessions motivating students to brainstorm themselves. <i>When you leave a hot tea for a while, it cools down. Why?</i> <i>Why do we feel hot when we place our hand near the fire/just air above the fire?</i> <i>Why does the water inside a vessel move continuously when it is on flame?</i> (Time allocated: 2-3 min.) - Students may come up with answers but they may not predict the phenomena exactly. (Time allocated: 3-4 min.) CLASS ACTIVITY: (20 min) - Facilitator summarises that transfer of heat is taking place and shows a PPT (Refer Annex II.1) on transfer of heat methods. (Time allocated: 18-20 min.) <i>Note: Facilitator has to explain in detail about the slides by pausing in the middle.</i> - Students are supposed to write key points and definitions while watching PPT. REINFORCEMENT: (8 min) - After watching the video, the facilitator poses the following questions and students in groups discuss among themselves and write answers in a notebook. (Time allocated: 2-3 min.) <i>In how many ways heat transfer can occur?</i> <i>Define conduction.</i> <i>What are meant by conductors of heat?</i> <i>What are meant by insulators of heat?</i> <i>Give some examples of conductors and insulators.</i> <i>Define convection.</i> <i>How does heat transfer take place in i) solids, ii) liquids and iii) gases?</i> <i>Define radiation.</i> <i>Sea breeze occurs during _____.</i> <i>Land breeze occurs during _____.</i>
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	6. After allotted time, the facilitator summarises the session by calling one member from each group to answer one question so that all groups get an opportunity. (Time allocated: 4-5 min.) (Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Creating)
Independent Practice (Homework)	Students will be asked to answer the Exercise Q. No. 8-11 in the textbook.
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	2 X 35 min.
Session Number	8 & 9	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Transfer of heat-activities
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to compare the transfer of heat in different methods while experimenting with activities.
Instructional Resource(s)	NCERT text book, required lab material, Task sheet-3 (Refer Annex IV.3)
Facilitation Procedure (How / Strategies/Activities/ Tools)	<u>SESSION-8:</u> JIGSAW STRATEGY is used to facilitate this session. CLASS ACTIVITY: (25 min) 1. Facilitator divides students into 4 groups of A, B, C and D and instructs them to open the textbook. (Time allocated: 2 min.) 2. Facilitator explains the main objectives of activities 3.6 and 3.7 and demonstrates how to do all activities and precautions.

	3. Now, students are asked to go through a textbook, read and discuss with group members. (Time allocated: 8-10 min.) 4. After the given time, the facilitator instructs all groups to do activities 3.6 and 3.7 and write observations in their notebook by writing each activity objective and observations separately. (Time allocated: 28-30 min.) REINFORCEMENT: (10 min) 5. Further, the facilitator calls one representative from each group to tell objectives and observations of that respective activity. (TELL) (Time allocated: 9-10 min.) Group A – activity 3.6, Group B – activity 3.7, 6. Facilitator summarises the session by collecting all observations randomly picking students to answer and correcting if necessary. (2-3 min.) (Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Creating)
Independent Practice (Homework)	Students will be asked to answer Task sheet-3 (Refer Annex IV.3)
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes
Facilitation Procedure (How / Strategies/Activities/ Tools)	<u>SESSION-9:</u> JIGSAW STRATEGY is used to facilitate this session.

	CLASS ACTIVITY: (25 min) 1. Facilitator divides students into 4 groups of A, B, C and D and instructs them to open the textbook. (Time allocated: 2 min.) 2. Facilitator explains the main objectives of activities 3.8 and 3.9 and demonstrates how to do all activities and precautions. (3. Now, students are asked to go through a textbook, read and discuss with group members. (Time allocated: 8-10 min.) 4. After the given time, the facilitator instructs all groups to do activities 3.8 and 3.9 and write observations in their notebook by writing each activity objective and observations separately. (Time allocated: 28-30 min.) REINFORCEMENT: (10 min) 5. Further, the facilitator calls one representative from each group to tell objectives and observations of that respective activity. (TELL) (Time allocated: 9-10 min.) Group C – activity 3.8 Group D – activity 3.9 6. Facilitator summarises the session by collecting all observations randomly picking students to answer and correcting if necessary. (2-3 min.) (Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Creating)
Independent Practice (Homework)	Students will answer the Exercise Q. No. 2, 3 & 4 from the textbook
Evaluation & Reflection	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

(Facilitator & Student reflection to be written by the Facilitator)	
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SESSION PLAN
SUBJECT- SCIENCE
CHAPTER -3 Heat

Class & Section	VII	Subject	Science
Estimated Number of Sessions	10	Duration of Session	35 min.
Session Number	10	Name of the Facilitator	

TOPIC	Heat
Sub-Topic	Kinds of clothes we wear in summer and winter
Essential Learning Outcomes (ELOs)	Students will be able to explain • The fundamentals of heat transfer mechanism in fluids and solids. • Applications of various heat transfer equipment. Students will be able to tell what type of clothes are preferable in winter and summer and why.
Instructional Resource(s)	NCERT text book, video-4 (Refer Annex III.4), worksheet-2 (Refer Annex I.2)
Facilitation Procedure (How / Strategies/Activities/ Tools)	READ ALOUD STRATEGY is used to facilitate this session. MOTIVATION: (7 min)

1. Facilitator initiates a session asking some prior knowledge questions to students.
 - a. *What type of clothes should we wear in summer and winter?*
 - b. *Why are you advised to use an umbrella when you go out in the sun?*
 - c. *Why do we prefer to wear light-coloured clothes in summer?*
 - d. *Why do we prefer to wear dark-coloured clothes in winter?*
 - e. *If you are given a choice of one thick blanket and two thick blankets joined together, which one would you choose and why?* (Time allocated: 2-3 min.)
2. Facilitator allows students to recall their prior knowledge individually. (RECALL)
3. Facilitator encourages students to come up with correct answers and randomly picks students to answer. (Time allocated: 3-4 min.)

CLASS ACTIVITY: (23 min)

4. Facilitator asks students to read activities 3.10 and activity 3.11 in the textbook. (Time allocated: 4-5 min.)
5. Now, the facilitator demonstrates activities 3.10 and 3.11 and instructs students to watch carefully and write observations in their notebook. (Time allocated: 9-10 min.)
Note: During activity 3.10, the facilitator can make use of flame to boil water instead of putting it under sunlight as it takes a lot of time.
6. Facilitator invites one student to read aloud all observations and continues the session concluding that dark-coloured things/clothes absorb more heat and vice-versa. (Time allocated: 5-6 min.)
7. Further, the facilitator plays a video-4 (Refer Annex III.4) on ‘why do woollen clothes keep us warm’ and students write important points. (Time allocated: 2 min.)

REINFORCEMENT: (5 min)

8. Facilitator summarises the session by revising all key points and activity conclusions and the preferences of wearing clothes during summer & winter. (Time allocated: 4-5 min.)

	(Bloom's Taxonomy Levels Targeted: Remembering, Understanding and Creating)
Independent Practice (Homework)	– Students will be able to answer worksheet-2 (Refer Annex I.2) – Students will answer the Exercise Q. No. 5, 6 & 7 from the textbook
Evaluation & Reflection (Facilitator & Student reflection to be written by the Facilitator)	To be Filled by the Facilitator at the end of the session to evaluate the Effective Learning Outcomes

Student Name: _____ **Grade:** 7**Subject:** Science**Chapter No. & Name:** 3. Heat**Date:** _____**Worksheet-1****Annexure:** I.1

Task: Apply your knowledge & understandability of concepts to answer following questions.

Instructions: Answer the following questions in a precise manner.

1. What do you mean by temperature?

Answer: _____

2. Which device is used to measure temperature?

Answer: _____

3. Name the silver stuff present in the thermometer?

Answer: _____

4. What is the unit of temperature adopted by India?

Answer: _____

5. What is the normal temperature of a human being?

Answer: _____

6. What is the range of a clinical thermometer?

Answer: _____

7. What is the best way to clean a thermometer?

Answer: _____

8. Why clinical thermometer cannot be used to measure high temperatures?

Answer: _____

9. What should be the level of mercury in the clinical thermometer before use?

Answer: _____

10. Why does the mercury not fall or rise in a clinical thermometer when taken out of the mouth?

Answer: _____

Learning outcome: Students will be able to apply their knowledge & understandability of concepts to answer above questions.

Student Name: _____ **Grade:** 7**Subject:** Science**Chapter No. & Name:** 3. Heat**Date:** _____**Worksheet-2****Annexure: I.2**

Task: Build your knowledge on transfer of heat and type of clothes we wear in summer and winter seasons while answering following questions.

Instructions: Read the questions carefully and answer in a precise manner.

1. How does heat flow from one object to another?

Answer: _____

2. What are conductors of heat?

Answer: _____

3. What are insulators of heat?

Answer: _____

4. How does heat transfer in liquids and gasses?

Answer: _____

5. Will heat transfer if the temperature between two objects is the same?

Answer: _____

6. How does heat transfer in liquids and gasses?

Answer: _____

7. How does heat transfer in solids?

Answer: _____

8. Does transfer of heat by radiation require any medium?

Answer: _____

9. Do all hot bodies radiate heat?

Answer: _____

10. Define: (a) Conduction, (b) convection and (c) radiation.

Answer: _____

Learning outcome: Build your knowledge on transfer of heat and type of clothes we wear in summer and winter seasons while answering following questions.

Student Name: _____ **Grade:** 7

Subject: Science

Chapter No. & Name: 3. Heat

Date: _____

Task Sheet-1
Annexure: IV.1

Task: Mark the below given objects as hot or cold.

Instructions: Put a tick mark for the objects whether it is cold or hot.

Sl. No.	Object	Cold/cool	Warm/hot
1.	TV which is running continuously for 6 hours		
2.	Ice cube		
3.	Tomato soup		
4.	Wooden handle of vessel		
5.	Water bottle in a fridge		

Task: Mark the tick below given objects as hot or cold.

Instructions: Which of the following materials do you think would be the best insulator to keep your hands warm on a cold winter day?

Sl. No.	Object	Cold/cool	Warm/hot
1.	Wool		
2.	Glass		
3.	Aluminum foil		
4.	Plastic		
5.	Cotton		
6.	Styrofoam		
7.	Metal		

Learning outcome: Students will be able to mark the objects as hot or cold by using their understandability of the concept.

Student Name: _____ Grade: 7

Subject: Science

Chapter No.& Name: 3. Heat

Date: _____

Task Sheet-2

Annexure: IV.2

Task: Answer the following based on temperature, measuring temperature and laboratory thermometer concepts.

Instructions: Read questions carefully

I. Fill in the blanks.

1. _____ is the degree of relative hotness or coldness.
2. Heat is a form of _____.
3. Mercury level falls easily in _____ thermometer.
4. SI unit of heat is _____.
5. Range of the clinical thermometer is from _____ °C _____ °C.
6. A kink is not provided in a _____ thermometer.

II. Name the following thermometers.

a. _____



b. _____



III. True or False.

1. Temperature of boiling water cannot be measured by a clinical thermometer.
(True/False)
2. We should hold the thermometer by the bulb while holding it. (True/False)
3. The bulb of the laboratory thermometer should not touch the bottom or sides of the container. (True/False)

IV. Multiple choice questions.

1. Name the device used for measuring temperature.
 - a. Thermometer
 - b. Ammeter
 - c. Spring balance
 - d. Voltmeter

Answer:

2. The unit of measurement of temperature is
 - a. Meter
 - b. Kilogram
 - c. Degree Celsius
 - d. Second

Answer:

3. The normal temperature of human body is
 - a. 40°C
 - b. 37°C
 - c. 42°C
 - d. 35°C

Answer:

4. The liquid metal used in a thermometer is

- a. Mercury
- b. Silver
- c. Gold
- d. Copper

Answer:

5. What is the use of kink in a thermometer?

- a. It prevents mercury level falling on its own
- b. It raises the mercury level
- c. It makes the thermometer look beautiful
- d. It acts as a joint to the glass tube that is used for mercury and the scale

Answer:

Learning outcome: Answer the following based on temperature, measuring temperature and laboratory thermometer concepts.

Student Name: _____ Grade: 7

Subject: Science

Chapter No. & Name: 3. Heat

Date: _____

Task Sheet-3

Annexure: IV.3

Task: Answer the following based on transfer of heat and kinds of clothes we wear in summer and winter concepts.

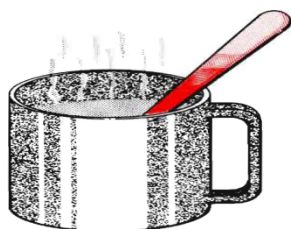
Instructions: Read the questions carefully.

I. Fill in the blanks.

1. Heat always flows from a hotter object to a _____ object.
2. Insulators are _____ conductors of heat.
3. Conductors are _____ conductors of heat.
4. From the sun, heat comes to us by the process called _____.
5. Land breeze blows at the time of _____.
6. Sea breeze blows at the time of _____.
7. Cotton or light-coloured clothes are preferable in _____ season.
8. Thick- coloured clothes are preferable in _____ season.

II. Name the following phenomena by using pictures given below.

1. _____.



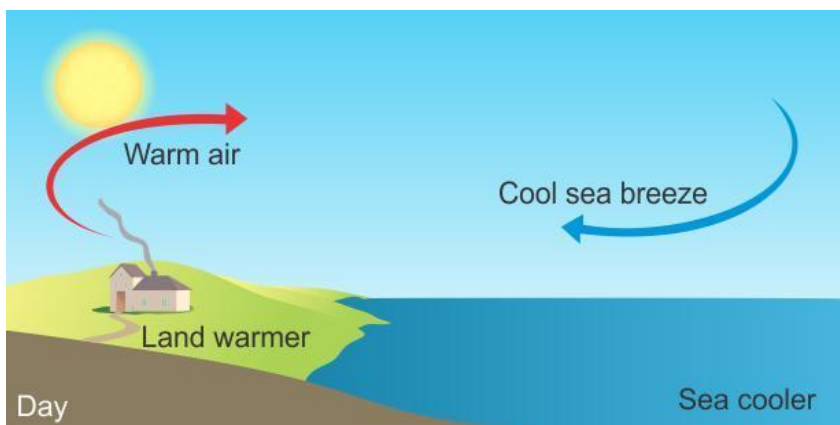
2. _____.



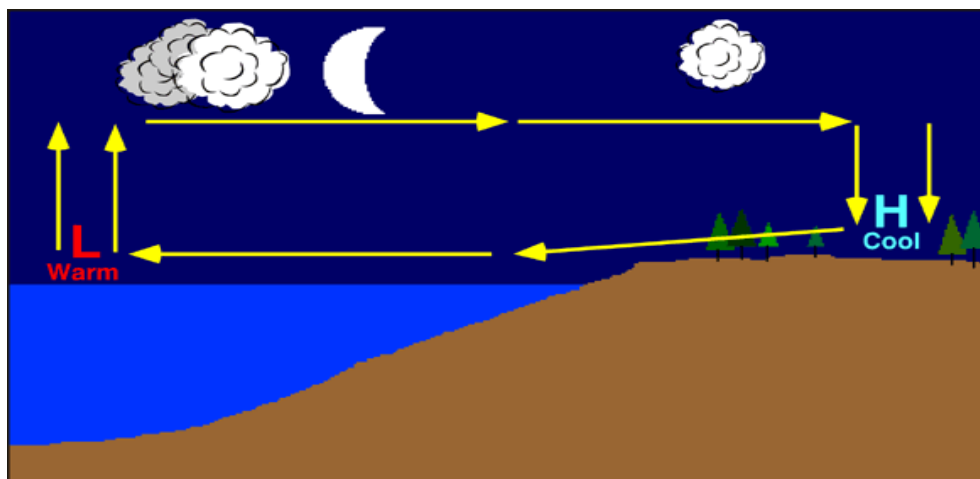
3. _____.



4. _____.



5. _____.



III. Multiple choice questions.

1. The maximum and minimum temperatures of a body are measured with a
 - a. Celsius thermometer
 - b. Fahrenheit thermometer
 - c. Maximum-minimum thermometer
 - d. Kelvin thermometer

2. The process by which heat flows from the hotter end to colder end of an object is called _____.
 - a. Conduction
 - b. Convection
 - c. Radiation
 - d. Vaporization

3. The process by which transfer of heat take place by the actual movement of particles from one part of the body to another part is called _____.
 - a. Conduction
 - b. Convection
 - c. Radiation
 - d. Vaporization

4. The transfer of heat which does not require any medium is called _____
 - a. Conduction
 - b. Convection
 - c. Radiation
 - d. Vaporization

5. The process by which heat comes from sun to earth is called _____
 - a. Conduction
 - b. Convection
 - c. Radiation
 - d. Vaporization

Learning outcome: Answer the following based on transfer of heat and kinds of clothes we wear in summer and winter concepts.